Validation and Calibration of the Remote Consolidated Approach of Reporting on Food Security (rCARI) Methodology

Preliminary results from the Central African Republic pilot study

Version 2.0

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Contents

Executive Summary

During the COVID-19 pandemic, a remote version of the Consolidated Approach for Reporting Food Insecurity (rCARI) was developed as an adaptation of the standard CARI methodology to report on food insecurity. The rCARI method differed from the standard CARI only in the economic vulnerability dimension. While the standard CARI used FES (food expenditure share) or ECMEN (economic capacity to meet essential needs) as an economic vulnerability indicator, rCARI used the household's income source and income change in place of FES because of the difficulty of collecting expenditure data over the phone. The rCARI methodology is has not yet been validated against the standard CARI.

Using a survey experiment methodology with stratified sample, this study aimed to evaluate and validate the rCARI methodology of collecting and reporting on food security against the standard CARI methodology. The sample frame for the study was the national population within which a random sample of households were selected in 2 out of the 20 prefectures (ADMIN1). A total of 2500 households were randomly selected and listed across 25 villages (primary sampling units-PSUs) and interviewed. The prefectures selected were done purposively based on mobile phone coverage, food security situation & accessibility (considering the security situation on the ground). Actual study respondents were randomly enlisted for the study through a random-walk methodology in each village (ADMIN5). Listed households were then randomly assigned to one of two study groups, the Face-to-Face (F2F) group, which received standard CARI modules in a F2F interview and Remote group which received phone interview (rCARI). In CAR, one of the key challenges was the low phone access (approximately 75% of listed households did not have access to mobile phones). To address this, 300 low-budget phones were randomly distributed to households that were randomly assigned to receive phone interviews but did not have access to phones.

The study interviewed a total of 1,245 household all together (969 F2F and 276 Remote interviews respectively).

This initial report is based on the analysis of data from the pilot study conducted in CAR and compares different versions of CARI computed using either remote or F2F modules. Three CARI version were calculated and compared with the standard CARI. These include: 1) CARI using FES from remote surveys, 2) CARI using income variable from F2F surveys, and 3) CARI using income sources and income change variable (the remote version of this indicator was implemented at the height of the Covid pandemic and thus for the purposes of this study will be referred to as standard rCARI).

The results show that;

- Households without mobile phones (usually excluded in mobile phone surveys) are significantly more food insecure (up to 35% more respondents) regardless of the CARI calculation methodology used.
- The aggregated CARI results show that, replacing the income source & income change indi-

cator with FES significantly underestimates food insecurity.

- Households, generally tend to report better food consumption and worse coping over the phone than they report in F2F.

1 Introduction

Food security assessment and reporting are at the core of WFP's mandate of addressing hunger and food insecurity. Over time, different methodologies (e.g., IPC, CARI, Cadre Harmonise, etc.) have been developed and used to assess and report global food insecurity each differing in scope, and purpose. A commonality that cuts across all methods is the use of primary household data collected through face-to-face interviews/surveys. However, in recent times particularly with the outbreak of the Covid-19 pandemic and attendant restrictions on movement and physical contact, data collection through conventional face-to-face method has been adversely affected. This impacted negatively the ability to assess food security needs because of the data gap created. Developing methodologies to accurately measure and interpret food insecurity especially in emergencies where physical access is restricted thus, is of great importance for WFP and many stakeholders. This is particularly important during emergencies when rapid assessment is required for advocacy to save lives and livelihoods but physical access to the field is restricted either because of health threats or security considerations for enumerators.

To leverage the increased penetration of technology in operational areas of WFP during the Covid lockdowns, a remote version of the Consolidated Approach for Reporting Food Insecurity (CARI)- the so-called rCARI- was developed as an adaptation of the standard CARI methodology and administered remotely (over the phone). The methodology though viewed favourably by researchers, and the larger humanitarian community, because of the stop-gap solution it provided during the covid-19 lockdown, questions about the validity of food security numbers it produces remain. For example, how are the remote modules comparable with already standardized face-to-face modules in capturing food insecurity. Although there exist validated studies on the use of face-to-face modules in assessing and reporting on food insecurity, remote methodology such as rCARI remains scant until now.

The potential advantages of the methodology abound for humanitarian work particularly for hard-to-reach areas. For example, the remote methodology has potential to save lives by providing real-time food insecurity data on the exposed population within the shortest possible time to inform policy. Furthermore it can result in better use of scarce resources to achieve high end results by rolling out few targeted modules to assess food insecurity of a larger population.

The overall objective of this study thus was to assess the validity of the remote consolidated approach for reporting food insecurity - rCARI module(s) against conventional CARI modules for measuring and reporting food insecurity. The study compared indicators collected through phone surveys with conventional face-to-face survey, considering the latter as the standard methodology. Specifically, food Consumption Score (FCS), reduced Coping Strategy Index (rCSI), Livelihood Coping Strategy (LCS), Food Expenditure Share (FES), and income sources and income change have been compared over the two modalities. The Economic Capacity to Meet Essential Needs (ECMEN) indicator which is sometimes used to aggregate CARI will not be tested in this study, as it requires the standard long expenditure module which cannot be collected over phone. In addition,

ECMEN calculation requires having an updated value of the minimum expenditure basket (MEB) of the study population, which is not available for the study areas. In its stead, a shortened version of the standard expenditure module was be used to test the possibility of collecting FES data over the phone.

Studying the validity of using the phone to collect food security indicators is a key step in judging the appropriateness of the rCARI methodology to report food insecurity prevalence. However, testing the comparability of these indicators through different modalities of data collection (Face-to-Face (F2F) vs. Phone) should be read within the wider consideration around phone survey modality of data collection. For example, other factors aside the modules should be considered while reading results of any remote survey.

1. Mobile phone ownership (mobile phone penetration rate in the study area) could likely exclude the most vulnerable population from the sample.
2. Mobile phone network coverage: Even in countries with high penetration rates, network coverage could be a problem, especially when electricity cuts are frequent and local infrastructure, such as communication masts, are not evenly distributed (World Bank, 2020).
3. Response rate: It is well documented that response rate in phone surveys is usually lower compared to F2F surveys (L'Engle et al., 2018; World Bank, 2020).

These factors may lead to skewed sample bias from phone surveys especially if the goal to report national representative results/outcomes. Nevertheless, there are several statistical weighting techniques that can adjust the results of phone surveys using baseline variables like using demographics characteristics of the most recent face-to-face surveys (Nagpal et al. 2021; Glazerman et al. 2020; Valliant et al. 2013; Elkasabi, 2015). This allows the results of such exercises to be computed and extrapolated to reflect some level of representativeness.

However, this current study may not be affected by these confounding issues since the main objective is not to report results at national level but to assess the effectiveness of remote modules to collect and report on food insecurity using a statistically comparable stratified sample. Thus, this current analysis does not employ weighting techniques to adjust for sampling skewness and response rate. Nonetheless, the study design could allow for testing of sample skewness due to mobile phone ownership, coverage and none response over the phone. CAR has one of the lowest mobile phone penetration rates globally (approximately 70% of the population do not own/use mobile phones).

2 Study Design & Sampling

The study was implemented as a survey experiment with stratified sample comparing the outcomes of two statistically comparable samples drawn from the same population.

Selection of study areas and participating households followed a stratified random sampling approach (stratified by the different Admin Areas-biggest to smallest but not representative at each admin area) based on 3-point criteria developed and agreed together with the WFP country office (CO) in Bangui. The criteria include, **1. Mobile phone coverage, 2. Food security situation, & 3. Accessibility (considering the security situation on the ground).** On the mobile phone coverage, a full list of areas with phone network was compiled through Orange/Moove/Telecel (the biggest mobile phone operators with nation wide coverage). Only Admin areas with phone coverage were included in the sampling frame to ensure only households with phone coverage had the probability of being listed for the study because of the remote interview part of the study.

The study design followed rigorous randomization procedures to assign potential study respondents to one of two study groups: F2F and Remote groups. The systematic steps in selecting the different administrative areas and study households with annotated explanation is presented in figure 1.

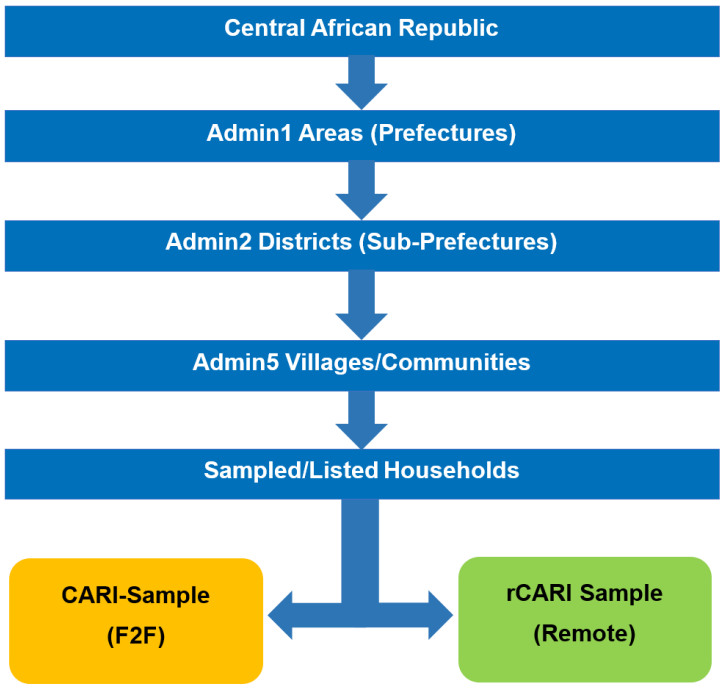

Figure 1: rCARI Validation Study Sampling Frame

1. STEP 1 - Based on the inclusion criteria (phone coverage, food security situation and accessibility), a list of prefectures meeting the compiled by the Country Office and two prefectures (Mbomou and Nana-Gribizi) were purposively selected as the study regions.

2. STEP 2 - Two sub-prefectures (ADMIN2), Bangassou and Kaga-Bandaro, were randomly selected from Mbomou and Nana-Gribizi Prefectures respectively. The Administrative structure of CAR allowed for inclusion of Admin3 areas (communes), therefore we randomly selected 4 communes from the 2 sub-prefectures. Bangassou, Nana, Botto, and Kanga-Bandoro were the 4 communes selected

3. STEP 3 - Selection of villages (ADMIN5): A list of villages in the selected communes - Nana, Botto, Kaga-Bandaro, and Bangassou that met the inclusion criteria set out in STEP 1 were compiled at this stage as primary sampling units (PSUs). A total of 25 PSUs were randomly selected (distributed uniformly across the selected communes) as sampling clusters to draw the households from. The selected villages took into consideration the possible variations between rural and urban food security. Therefore, the final PSUs was made up of 15 urban and 10 rural villages respectively. See annex for the full list of administrative areas selected, number of households listed and the phone ownership. Note: the total sample presented are slightly different because 420 households listed without network coverage were dropped before randomization).

4. STEP 4 - Enlisting households for the study: The sampling frame for study was all households in selected PSUs. A minimum of 100 households per PSU were selected through the random walk approach (listing every 3rd house in the sequence) by trained enumerators. Consent of participation was taken from all HHs before being included into the study sample (a total of 2,500 HHs). The final listed households in this stage were used for the randomization process in the next step.

5. STEP 5 - Random assignment of household (hh): Consented final sampled households in STEP 4 are randomly assigned into one of two survey modalities/groups; F2F (CARI group) and Remote (rCARI group). The experiment was designed such that the full sample was divided equally between the two modalities.

6. STEP 6 - Mobile phone assignment to households without phones. Give the low mobile phone penetration in CAR (more than 70% households are without access to phones), low budget phones were for 300 households without phones but randomy assigned to receive phone interviews (detail description of this process is given in section 2.2).

   All interviews were conducted over the same time frame to ensure that households in the two modalities are reporting their food security situation over the same window.

2.1 Sampling survey respondents & survey implementation

The household sampling frame for the study was all households in the 25 villages (PSUs) selected. Approximately 100 households per PSU were randomly listed (after receiving their informed consent) for the study. The final isted for the study before randomization was 2500 households. A number of baseline demographic and household level characteristics (such as age, gender education

status of the household head, household size, and ownership of common assets etc.) were collected during listing phase. The contact details (address and phone number if any) of the listed households were also collected during the listing.

For the data collection, enumerators were trained by in-person RAM facilitators from WPF-HQ with support from RBD (Dakar) and the VAM country office team in CAR. Enumerator training was done in French over a 3-day period with one additional day used for field testing, adjusting, and adapting the final questionnaires (F2F & Remote).

Interviews (both F2F & Remote) were conducted in local languages of each locality. Enumerator were sent to localities where they could speak the local dialects.

2.2 Phone assignment for remote modality

In CAR, as discussed earlier, not only is the phone connectivity an issue, but phone ownership is generally low (only 27% own phones from the national statistics). This implies that, a larger proportion of the population i.e., a little over 70% of households (usually lower income earners and likely to be food insecure) are automatically excluded from phone survey sample frame.

The situation in CAR thus presented a unique opportunity to investigate deeper to understand the type of households/respondents are usually excluded in such exercises, their food security situation and how they cope with shocks or stress associated with food insecurity.

In this regard, a budget line was created to provide small hand-held phones to a sub-sample of the listed households (households with network coverage but without mobile phones but assigned randomly to the remote group). The budget was for 300 phones, however since the final sample without phones was more than budgeted, 300 households/respondents were randomly selected and given pre-connected phones with SIM cards (refer to Figure 1 for the different levels of randomization was implemented in the study and the corresponding sample size at each level).

A number of strategies were put in place to ensure the phones distribution achieved high success. First, all mobile phones were fully charged and fixed with operational SIM cards before being distributed to selected households/respondents (as a way to address electricity access in rural areas and operation challenges associated with the new phone respectively). Second, there was coordination between the field team distributing the phones and the team conducting the phone interviews in Bangui to ensure that calls were made to households/respondents immediately they received the mobile phones from the field team.

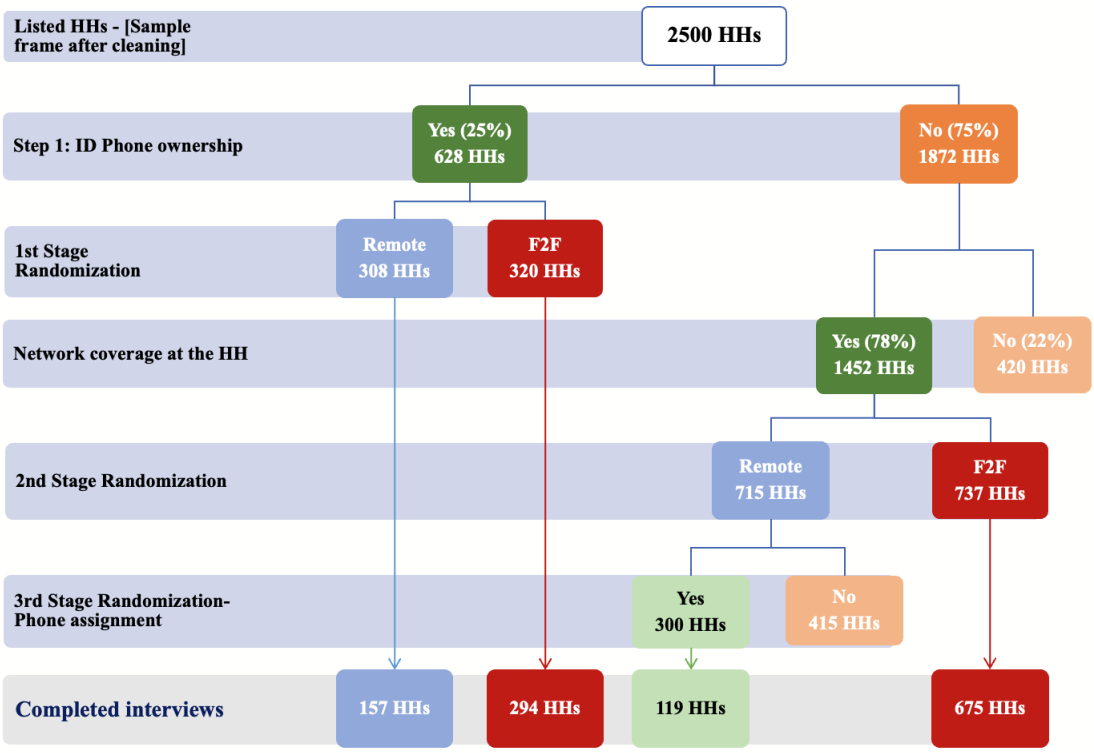

Figure 2: Sample Randomization and Survey Completion

Data collection was conducted from July - August 2023 in the selected study areas using trained enumerators from the statistical bureau of CAR-ICASEES. In the next section, the baseline comparison of the different groups is presented using the observable characteristics collected during the household listing phase.

3 Descriptive statistics

In this section, the core descriptive statistics of the study sample is presented. Given the rather elaborate design that has been implemented, a number of statistics in the form of interview completion, balance tables and attrition statistics across the different level of implementation are presented.

3.1 Survey completion

Table 2 shows the interview completion rate between F2F and Remote modalities. In total, the survey was able to complete 60% (1,245 interviews) of the original sample with the overall completion rate higher in the F2F (92%) compared to the Remote modality (27%).

Table 1: Sample Completion: Face-to-Face vs. Remote

	Not Completed		Completed		Total	
	Count	Percent	Count	Percent	Count	Percent
Face-to-Face	88	8.33	969	91.67	1057	100.00
Remote	747	73.02	276	26.98	1023	100.00
Total	835	40.14	1245	59.86	2080	100.00

The study sample was sub-categorized into phone-owners and non-phone-owners based on the study design to capture phone and network coverage.

3.2 Randomization quality check

One of the cardinal requirements of the study design is the statistical comparability of the two study groups at the time of data collection (balance test) based on observable characteristics. To assess the statistical comparability (equivalence) of the two study groups - F2F and Remote - various baseline demographics, and other observable characteristics collected during the listing phase have been compared using the appropriate statistical tools. This helped to evaluate the equivalence of groups across the different levels of randomization implemented in the study. Specifically, the balance test was done for the following groups: 1) modality assignment (F2F & Remote), and 2) module length assignment (Long & Short).

3.2.1 Modality randomization

The statistical balance test for modality assignment is presented here. The sampled households have been categorized into tree different sub-samples and tested for their statistical comparability based on random assignment to the two study groups to receive F2F or Remote interviews. The three sub-sample are: 1) Phone-owners, 2) No phone sample, and 3) the full sample. The full description of these sub-samples is given in Figure 1.

The balance test results show that based on the baseline characteristics, the randomization achieved a good statistical balance, i.e., households random assigned to the study groups were statistically similar/equivalent.

Table 2: Balance Table on Randomized Survey Modality Assignment: Face-to-Face vs. Remote

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
List respondent age	35.97	37.59	1.61	39.42	39.35	-0.08	38.38	38.82	0.44
	(12.42)	(12.76)	(0.11)	(13.95)	(14.24)	(0.92)	(13.59)	(13.83)	(0.47)
Household head is the respondent	0.72	0.73	0.01	0.78	0.79	0.02	0.76	0.78	0.02
	(0.45)	(0.44)	(0.68)	(0.42)	(0.40)	(0.43)	(0.43)	(0.42)	(0.38)
Female Household Head	0.20	0.23	0.03	0.30	0.31	0.01	0.27	0.28	0.02
	(0.40)	(0.42)	(0.30)	(0.46)	(0.46)	(0.66)	(0.44)	(0.45)	(0.37)
Married Household Head	0.25	0.26	0.01	0.14	0.11	-0.03	0.17	0.16	-0.02
	(0.44)	(0.44)	(0.85)	(0.35)	(0.31)	(0.09)	(0.38)	(0.36)	(0.25)
Household head years of education	7.50	7.49	-0.02	3.39	3.30	-0.08	4.63	4.56	-0.07
	(4.10)	(4.50)	(0.96)	(3.27)	(3.22)	(0.63)	(4.01)	(4.12)	(0.70)
Displaced household	0.11	0.12	0.01	0.03	0.03	-0.00	0.06	0.06	0.00
	(0.31)	(0.32)	(0.67)	(0.18)	(0.18)	(0.85)	(0.23)	(0.23)	(0.86)
Household size	7.68	7.95	0.28	6.09	5.96	-0.13	6.57	6.56	-0.01
	(3.50)	(3.95)	(0.35)	(2.66)	(2.65)	(0.37)	(3.03)	(3.23)	(0.96)
Number of household member younger than 15	4.02	4.00	-0.02	3.21	3.13	-0.08	3.45	3.39	-0.06
	(2.27)	(2.12)	(0.93)	(1.92)	(1.90)	(0.41)	(2.06)	(2.01)	(0.47)
Number of household member older than 60	0.21	0.22	0.01	0.18	0.18	0.00	0.19	0.19	0.01
	(0.48)	(0.55)	(0.72)	(0.44)	(0.48)	(0.87)	(0.45)	(0.50)	(0.73)
Has household member with disability/chronic disease	0.29	0.30	0.00	0.15	0.16	0.01	0.19	0.20	0.01
	(0.46)	(0.46)	(0.96)	(0.36)	(0.37)	(0.50)	(0.40)	(0.40)	(0.59)
Number of rooms	2.84	2.99	0.15	1.96	1.91	-0.05	2.23	2.24	0.01
	(1.25)	(1.34)	(0.14)	(1.08)	(1.09)	(0.40)	(1.20)	(1.27)	(0.83)
Household Crowding Index	3.08	3.00	-0.08	4.02	4.00	-0.02	3.74	3.70	-0.03
	(1.79)	(1.64)	(0.57)	(2.94)	(2.97)	(0.92)	(2.68)	(2.68)	(0.78)
Household has impoverished roof	0.67	0.66	-0.01	0.85	0.86	0.01	0.79	0.80	0.00
	(0.47)	(0.48)	(0.80)	(0.36)	(0.35)	(0.57)	(0.41)	(0.40)	(0.79)
Household has impoverished floor	0.89	0.91	0.02	0.96	0.96	0.00	0.94	0.95	0.01
	(0.32)	(0.29)	(0.37)	(0.19)	(0.19)	(0.87)	(0.24)	(0.22)	(0.44)
Observations	320	308	628	737	715	1,452	1,057	1,023	2,080

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

This means that, the outcome of in this case, food security would likely be comparable between the groups holding all things constant and accounting for attrition.

3.2.2 Questionnaire length randomization-F2F

As described in the study set-up, aside from households being randomly assigned to received F2F and Remote, households were further randomized to receive either short- or long-modules of the FES, FCS and LCS. This section presents the balance test for this second level of randomization that was implemented in the study. Table 4 presents the balance test for the F2F group while Table 5 presents that of the Remote sub-group respectively. Both Table 3 & 4 show that statistically, household assignment to the short- and long-modules was successfully balanced.

Given that results of the randomization was balanced for baseline observables, the main results of study showing the different outcome indicators calculated are presented in the next section (for the results of the post survey attrition analysis, refer to the attrition tables in the Annexes).

Table 3: Balance Table for Randomized Questionnaire Length Assignment in F2F Survey: Long vs. Short

	Own Phone			No Phone			Total F2F		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
List respondent age	35.72	36.24	0.52	40.13	38.66	-1.47	38.81	37.92	-0.89
	(12.73)	(12.11)	(0.71)	(14.66)	(13.11)	(0.15)	(14.24)	(12.85)	(0.29)
Household head is the respondent	0.71	0.72	0.00	0.80	0.76	-0.04	0.77	0.75	-0.03
	(0.45)	(0.45)	(0.93)	(0.40)	(0.43)	(0.22)	(0.42)	(0.44)	(0.33)
Female Household Head	0.20	0.19	-0.01	0.30	0.30	-0.00	0.27	0.26	-0.01
	(0.40)	(0.40)	(0.84)	(0.46)	(0.46)	(0.94)	(0.44)	(0.44)	(0.85)
Married Household Head	0.24	0.26	0.02	0.12	0.16	0.03	0.16	0.19	0.03
	(0.43)	(0.44)	(0.70)	(0.33)	(0.37)	(0.18)	(0.37)	(0.39)	(0.19)
Household head years of education	7.30	7.71	0.41	3.32	3.46	0.13	4.52	4.76	0.24
	(4.04)	(4.16)	(0.38)	(3.23)	(3.32)	(0.58)	(3.94)	(4.09)	(0.34)
Displaced household	0.13	0.08	-0.06	0.04	0.02	-0.02	0.07	0.04	-0.03*
	(0.34)	(0.27)	(0.10)	(0.21)	(0.15)	(0.10)	(0.26)	(0.19)	(0.02)
Household size	7.46	7.91	0.45	6.17	6.00	-0.17	6.56	6.58	0.03
	(3.35)	(3.65)	(0.25)	(2.82)	(2.47)	(0.38)	(3.05)	(3.01)	(0.89)
Number of household member younger than 15	3.87	4.17	0.30	3.28	3.13	-0.15	3.46	3.45	-0.01
	(2.13)	(2.41)	(0.24)	(2.07)	(1.74)	(0.29)	(2.10)	(2.02)	(0.95)
Number of household member older than 60	0.21	0.21	0.00	0.19	0.16	-0.03	0.20	0.17	-0.02
	(0.49)	(0.47)	(0.94)	(0.43)	(0.45)	(0.30)	(0.45)	(0.45)	(0.43)
Has household member with disability/chronic disease	0.32	0.27	-0.05	0.16	0.14	-0.01	0.21	0.18	-0.02
	(0.47)	(0.44)	(0.35)	(0.36)	(0.35)	(0.61)	(0.40)	(0.39)	(0.34)
Number of rooms	2.78	2.90	0.12	2.02	1.89	-0.13	2.25	2.20	-0.05
	(1.25)	(1.24)	(0.38)	(1.10)	(1.05)	(0.10)	(1.20)	(1.20)	(0.50)
Household Crowding Index	3.04	3.13	0.08	3.88	4.17	0.29	3.63	3.85	0.22
	(1.71)	(1.89)	(0.67)	(2.76)	(3.13)	(0.18)	(2.52)	(2.85)	(0.18)
Household has impoverished roof	0.70	0.63	-0.06	0.84	0.86	0.02	0.79	0.79	-0.00
	(0.46)	(0.48)	(0.25)	(0.37)	(0.35)	(0.36)	(0.41)	(0.41)	(0.92)
Household has impoverished floor	0.92	0.85	-0.07	0.96	0.96	0.00	0.95	0.93	-0.02
	(0.27)	(0.36)	(0.06)	(0.19)	(0.19)	(0.85)	(0.22)	(0.26)	(0.19)
Observations	164	156	320	382	355	737	546	511	1,057

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 4: Balance Table for Randomized Questionnaire Length Assignment in Remote Survey: Long vs. Short

	Own Phone			No Phone			Total Remote		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
List respondent age	37.14	38.07	0.93	39.42	39.28	-0.14	38.71	38.93	0.22
	(13.19)	(12.29)	(0.52)	(14.01)	(14.48)	(0.90)	(13.79)	(13.88)	(0.80)
Household head is the respondent	0.75	0.71	-0.03	0.81	0.78	-0.03	0.79	0.76	-0.03
	(0.44)	(0.45)	(0.54)	(0.39)	(0.42)	(0.30)	(0.41)	(0.43)	(0.25)
Female Household Head	0.20	0.27	0.07	0.31	0.31	-0.00	0.28	0.29	0.02
	(0.40)	(0.44)	(0.17)	(0.46)	(0.46)	(0.89)	(0.45)	(0.46)	(0.52)
Married Household Head	0.23	0.29	0.06	0.11	0.11	-0.00	0.15	0.16	0.01
	(0.42)	(0.46)	(0.21)	(0.32)	(0.31)	(0.92)	(0.36)	(0.37)	(0.53)
Household head years of education	7.69	7.27	-0.42	3.51	3.10	-0.41	4.81	4.31	-0.49
	(4.35)	(4.66)	(0.41)	(3.31)	(3.13)	(0.09)	(4.14)	(4.10)	(0.06)
Displaced household	0.11	0.12	0.01	0.04	0.03	-0.01	0.06	0.05	-0.01
	(0.32)	(0.33)	(0.77)	(0.19)	(0.16)	(0.29)	(0.24)	(0.23)	(0.58)
Household size	7.99	7.92	-0.07	5.91	6.02	0.11	6.55	6.57	0.02
	(3.30)	(4.57)	(0.88)	(2.79)	(2.51)	(0.58)	(3.11)	(3.36)	(0.93)
Number of household member younger than 15	3.96	4.05	0.08	3.08	3.17	0.08	3.36	3.42	0.07
	(2.01)	(2.25)	(0.73)	(2.02)	(1.76)	(0.55)	(2.06)	(1.95)	(0.59)
Number of household member older than 60	0.24	0.20	-0.04	0.14	0.22	0.07*	0.17	0.21	0.04
	(0.60)	(0.48)	(0.54)	(0.40)	(0.55)	(0.04)	(0.47)	(0.53)	(0.22)
Has household member with disability/chronic disease	0.24	0.36	0.12*	0.18	0.15	-0.04	0.20	0.21	0.01
	(0.43)	(0.48)	(0.02)	(0.39)	(0.35)	(0.19)	(0.40)	(0.41)	(0.70)
Number of rooms	3.04	2.95	-0.09	1.89	1.94	0.06	2.24	2.23	-0.01
	(1.39)	(1.29)	(0.55)	(1.10)	(1.07)	(0.49)	(1.31)	(1.23)	(0.91)
Household Crowding Index	2.95	3.06	0.11	4.07	3.94	-0.13	3.72	3.68	-0.04
	(1.29)	(1.96)	(0.56)	(3.09)	(2.84)	(0.55)	(2.71)	(2.65)	(0.81)
Household has impoverished roof	0.66	0.65	-0.01	0.85	0.86	0.01	0.79	0.80	0.01
	(0.48)	(0.48)	(0.92)	(0.36)	(0.34)	(0.68)	(0.41)	(0.40)	(0.70)
Household has impoverished floor	0.94	0.88	-0.06	0.97	0.96	-0.01	0.96	0.94	-0.02
	(0.24)	(0.33)	(0.07)	(0.18)	(0.19)	(0.68)	(0.20)	(0.24)	(0.13)
Observations	161	147	308	358	357	715	519	504	1,023

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4 Results and discussions

In this section, the main results of the study aggregated by the food security indicator are analysed and presented. Here, the different versions of CARI calculated using remote or F2F is compared with the standard CARI. A test of group means using student t-test of equivalence is performed. The comparison is at the sub-sample level of phone-ownership, no phone, and the combined full sample.

4.1 CARI vs. rCARI

Conventional/standard CARI: CARI-FES: Standard CARI using FES from F2F surveys.

Other ways of aggregating CARI (that are tested)

- rCARI-FES: Standard CARI using FES from remote surveys.
- CARI-Inc: CARI using the income variable from the F2F surveys.
- rCARI-Inc: Standard rCARI using income and income change variable from remote surveys (this is the CARI version that was used during COVID 19 Pandemic).

The different methods of calculating CARI that are tested against the standard/conventional CARI are shown in Table 6.

Table 5: Lists the different aggregation methods and indicators used for each methodology

	Aggregation Methodology			
Indicator	**CARI (F2F)**	**CARI (F2F)**	**rCARI (Remote)**	**rCARI (Remote)**
FCS	Yes	Yes	Yes	Yes
FES	Yes	**No**	Yes	No
LCS	Yes	Yes	Yes	Yes
rCSI	Yes	Yes	Yes	Yes
Income source & income change	**No**	Yes	**No**	**Yes**

For more information on the calculation of food security indicators included in each methodology refer to Annexes attached.

The first comparison is between CARI aggregated using remote modules and replacing FES with income source and income change variable (rCARI-Inc). The results are presented Table 7 is for all the 3 sub-sample comparisons.

The main significant finding across the results in Table 7 is that rCARI-Inc, significantly under-reports the number of food insecure people. The difference is found to be 21 percentage points amongst phone-owners sub-sample, 9 percentage points in the no-phone group and up to 23 percentage points in the combined full sample. This means that, the income source in its current form (as was used during the covid pandemic) likely underestimated the number of people that were food insecure compared to what would have been found if the standard CARI-FES was implemented around the same time.

Table 6: Impact of Survey Modality: Standard CARI-FES vs. Standard rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.01	0.01	-0.00	0.00	0.02	0.02	0.00	0.01	0.01
	(0.08)	(0.08)	(0.96)	(0.00)	(0.13)	(0.15)	(0.05)	(0.10)	(0.17)
Marginally Food Secure	0.31	0.52	0.21***	0.07	0.14	0.07*	0.14	0.36	0.22***
	(0.46)	(0.50)	(0.00)	(0.25)	(0.35)	(0.03)	(0.35)	(0.48)	(0.00)
Moderately Food Insecure	0.60	0.45	-0.15**	0.74	0.70	-0.04	0.70	0.56	-0.14***
	(0.49)	(0.50)	(0.00)	(0.44)	(0.46)	(0.39)	(0.46)	(0.50)	(0.00)
Severely Food Insecure	0.08	0.02	-0.06**	0.20	0.14	-0.05	0.16	0.07	-0.09***
	(0.27)	(0.14)	(0.00)	(0.40)	(0.35)	(0.14)	(0.37)	(0.26)	(0.00)
Food Insecure Group	0.68	0.47	-0.21***	0.93	0.84	-0.09**	0.86	0.63	-0.23***
	(0.47)	(0.50)	(0.00)	(0.25)	(0.37)	(0.01)	(0.35)	(0.48)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

While on the overage the remote modality slightly underestimates the number overall food insecure, across the sub-samples compared to F2F, comparing the non-phone owners with phone-owners, there are between 25 and 37 percent more food insecure people estimated in the F2F and Remote respectively in non-phone owners. Given that the research objectives seeks to compare the results of the standard F2F with remote surveys, it is important to compare the results of the F2F results from the full sample with the remote results from the phone owners sub-sample. The result of this comparison shows that using the remote methodology in its current setup using the income source and income change indicators significantly underreports acute food insecurity by 39 percentage points.

The next level of comparison is the CARI computed using FES from remote surveys with the standard CARI modules presented in Table 8. The results show different patterns across the 3 groups. First, for the phone-owners although the rCARI-FES in terms of absolute terms produces higher number of food insecure people, the difference is not statistically significantly compared to the standard CARI-FES. Second, for the no phone owners and the full sample, rCARI-FES significantly under reports the food insecure numbers by between 7 and 8 percentage points. The overall effect is driven by the moderately food insecure category where rCARI-FES significantly under reports the numbers by 15 and 11 percentage points respectively for none phone owners and full-sample.

The results of the comparison in Table 8 shows that using the FES module in phone surveys will likely result in under-reporting food insecurity compared to the standard F2F FES module.

A cursory analysis of the results in Table 8 shows that (as expected) the none phone owners are significantly more food insecure (moderately food insecure and severely food insecure). Although using FES indicator slightly overestimates the number of food insecure people by between 68 and 71 percent in F2F and Remote respectively in the phone-owners sub-sample, the numbers are 93 and 87 percent respectively amongst the none phone owners.

In the next analysis, CARI computed using income source and income change variable is compared over the modality. The same module asked over phone is compared with the F2F responses. The

Table 7: Impact of Survey Modality: CARI-FES vs. rCARI-FES

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.01	0.01	-0.00	0.00	0.01	0.01	0.00	0.01	0.01
	(0.08)	(0.08)	(0.96)	(0.00)	(0.09)	(0.32)	(0.05)	(0.08)	(0.33)
Marginally Food Secure	0.31	0.29	-0.02	0.07	0.13	0.06	0.14	0.22	0.08**
	(0.46)	(0.45)	(0.61)	(0.25)	(0.33)	(0.07)	(0.35)	(0.41)	(0.01)
Moderately Food Insecure	0.60	0.58	-0.02	0.74	0.59	-0.15**	0.70	0.58	-0.11***
	(0.49)	(0.50)	(0.65)	(0.44)	(0.49)	(0.00)	(0.46)	(0.49)	(0.00)
Severely Food Insecure	0.08	0.13	0.05	0.20	0.28	0.08	0.16	0.19	0.03
	(0.27)	(0.33)	(0.14)	(0.40)	(0.45)	(0.06)	(0.37)	(0.39)	(0.24)
Food Insecure Group	0.68	0.71	0.02	0.93	0.87	-0.07*	0.86	0.78	-0.08**
	(0.47)	(0.46)	(0.61)	(0.25)	(0.34)	(0.04)	(0.35)	(0.42)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

result of this comparison is presented in Table 9.

The results show that, amongst the phone-owners sample, there is no statistical difference in the food insecurity outcomes between the remote and F2F. However, amongst the none phone owners group, the income change indicator significantly over states the number of severely food insecure by 11 percentage points over the phone (classifying the overall food insecure by 8 percentage points) compared to the F2F module. The results looking at the combined sample show that though the remote version shows a bi-directional effect underestimating the number of moderately food insecure (8%) and overestimating the severely food insecure (4%).

Table 8: Impact of Survey Modality: CARI-Inc vs. rCARI-Inc

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.02	0.01	-0.01	0.00	0.02	0.02	0.01	0.01	0.00
	(0.14)	(0.08)	(0.18)	(0.04)	(0.13)	(0.20)	(0.08)	(0.10)	(0.59)
Marginally Food Secure	0.51	0.52	0.01	0.24	0.14	-0.09**	0.32	0.36	0.04
	(0.50)	(0.50)	(0.86)	(0.43)	(0.35)	(0.01)	(0.47)	(0.48)	(0.25)
Moderately Food Insecure	0.45	0.45	0.01	0.72	0.70	-0.03	0.64	0.56	-0.08*
	(0.50)	(0.50)	(0.89)	(0.45)	(0.46)	(0.55)	(0.48)	(0.50)	(0.02)
Severely Food Insecure	0.02	0.02	-0.00	0.04	0.14	0.11**	0.03	0.07	0.04*
	(0.14)	(0.14)	(0.92)	(0.19)	(0.35)	(0.00)	(0.18)	(0.26)	(0.01)
Food Insecure Group	0.47	0.47	0.01	0.76	0.84	0.08*	0.67	0.63	-0.04
	(0.50)	(0.50)	(0.91)	(0.43)	(0.37)	(0.04)	(0.47)	(0.48)	(0.21)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

The next level of analysis looks at the food security outcomes using a paired sample t-test of means. Here, CARI is calculated for the same household/respondent using FES and Income source and income change separately. Table 10 presents the comparison for the paired F2F sample while Table 11 presents that for the remote sample.

Table 10 show that the income indicator significantly under-reports food insecurity by between 17 and 22 percentage points across all sub-samples. In terms of the none-phone owners and phone owners, the difference is much pronounced (about 30 percentage more using the income indicator and 25 percentage points using the FES: See Table 10).

Table 9: Impact of Indicator Composition: Paired CARI-FES vs. CARI-Inc in F2F Surveys

	Own Phone			No Phone			Total		
Food Security Category	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff
Food Secure	0.01	0.02	0.01**	0.00	0.00	0.00	0.00	0.01	0.01*
	(0.00)	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Marginally Food Secure	0.31	0.51	0.2***	0.07	0.24	0.17***	0.14	0.32	0.18***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)
Moderately Food Insecure	0.60	0.45	-0.16***	0.74	0.72	-0.01	0.70	0.64	-0.06**
	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.01)	(0.02)	(0.02)
Severely Food Insecure	0.08	0.02	-0.06***	0.20	0.04	-0.16***	0.16	0.03	-0.13***
	(0.02)	(0.01)	(0.02)	(0.02)	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)
Food Insecure Group	0.68	0.47	-0.22***	0.93	0.76	-0.17***	0.86	0.67	-0.18***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)
Observations	294	294	294	675	675	675	969	969	969

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

In Table 11, the results still show that the income indicator households/respondents tend to significantly under-report their food insecurity using the income indicator (24% for phone owners & 14% for the combined sample). Although the difference is significantly different amongst the none-phone owners sub-sample, it can be observed that using the income indicator respondents regardless of the modality to under-report their food insecurity compared to using FES module.

Table 10: Impact of Indicator Composition: Paired CARI-FES vs. CARI-Inc in Remote Surveys

	Own Phone			No Phone			Total		
Food Security Category	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff
Food Secure	0.01	0.01	0.00	0.01	0.02	0.01	0.01	0.01	0.00
	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)
Marginally Food Secure	0.29	0.52	0.24***	0.13	0.14	0.02	0.22	0.36	0.14***
	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.02)	(0.03)	(0.03)
Moderately Food Insecure	0.58	0.45	-0.13*	0.59	0.70	0.11*	0.58	0.56	-0.03
	(0.04)	(0.04)	(0.05)	(0.05)	(0.04)	(0.05)	(0.03)	(0.03)	(0.04)
Severely Food Insecure	0.13	0.02	-0.11***	0.28	0.14	-0.13**	0.19	0.07	-0.12***
	(0.03)	(0.01)	(0.03)	(0.04)	(0.03)	(0.04)	(0.02)	(0.02)	(0.02)
Food Insecure Group	0.71	0.47	-0.24***	0.87	0.84	-0.03	0.78	0.63	-0.14***
	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Observations	157	157	157	119	119	119	276	276	276

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

To conclude on the CARI and rCARI comparisons, three important highlights are observed. First, the sample that did not own phones (none phone owners sub-sample) are consistently more food insecure regardless of which modality/module is used to collect the data (76% food insecure in F2F & 84% in remote modality). Thus, in a typical CATI (computer assisted telephone interviews) surveys using random digit dialling where only a sub-population of respondents with access to phones are interviewed, it automatically exclude the non phone owners who are likely to be more food insecure and vulnerable. Second, the remote CARI methodology regardless of the indicators used to compute AFI significantly under-reports AFI. This is found to be between 15 and 39 percentage points. This means that people tend to generally under-report their food insecurity over the phone. Third, using the income source and income change indicator instead of the FES significantly over reports food security regardless of the modality (F2F or remote) used in collecting it.

4.2 Comparison of the food security indicators across modality

In this section, the results of the individual indicators compared over the two modalities (F2F vs remote) are compared statistically and presented.

4.2.1 Food Consumption Score (FCS)

The Food Consumption Score (FCS) is a composite score based on households' dietary diversity, food consumption frequency, and relative nutritional importance of different food groups, and is calculated by inspecting how often households consume food items from the different food groups during a 7-day reference period. The responses are aggregated, and weights applied to obtain an index which is then used to categorize the household into one of three groups of food consumption: food secure, borderline, and poor.

The results show that consistently, the remote FCS module significantly overreported food security (reporting better food consumption) compared to the F2F module; 8 points amongst the phone owners sub-sample, 4 & 10 points more amongst the none phone owners and the full sample respectively (see Table 12). These difference between the F2F and remote responses is driven by higher consumption of tubers and cereals, protein-rich foods, dairy products, vegetables, and sugar over the phone.

In terms of the three FCS Groups, consistently, the remote module under reports the number of households with poor food consumption by between 12 and 23 percentage points.

Table 11: Impact of Survey Modality on Food Consumption Score and Groups

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Consumption Score	39.76	47.63	7.87***	24.59	28.12	3.53**	29.19	39.22	10.03***
	(17.58)	(17.45)	(0.00)	(12.67)	(12.00)	(0.00)	(15.94)	(18.12)	(0.00)
FCS: Acceptable	0.40	0.58	0.18***	0.10	0.09	-0.01	0.19	0.37	0.18***
	(0.49)	(0.50)	(0.00)	(0.31)	(0.29)	(0.70)	(0.40)	(0.48)	(0.00)
FCS: Borderline	0.35	0.29	-0.06	0.24	0.39	0.15**	0.27	0.33	0.06
	(0.48)	(0.45)	(0.19)	(0.43)	(0.49)	(0.00)	(0.45)	(0.47)	(0.07)
FCS: Poor	0.25	0.13	-0.12**	0.66	0.52	-0.14**	0.53	0.30	-0.23***
	(0.43)	(0.34)	(0.00)	(0.48)	(0.50)	(0.01)	(0.50)	(0.46)	(0.00)
Consumption over the past 7 days (cereals and tubers)	5.87	6.24	0.36*	5.08	4.87	-0.21	5.32	5.64	0.33*
	(1.86)	(1.33)	(0.02)	(2.17)	(2.13)	(0.31)	(2.11)	(1.85)	(0.01)
Consumption over the past 7 days (pulses)	3.74	3.74	-0.01	2.01	1.98	-0.03	2.54	2.98	0.45**
	(2.33)	(2.08)	(0.98)	(1.92)	(1.62)	(0.88)	(2.20)	(2.08)	(0.00)
Consumption over the past 7 days (dairy products)	0.50	1.14	0.64**	0.07	0.40	0.33**	0.20	0.82	0.62***
	(1.57)	(2.27)	(0.00)	(0.55)	(1.11)	(0.00)	(0.99)	(1.90)	(0.00)
Consumption over the past 7 days (protein-rich foods)	1.72	2.09	0.37*	0.51	0.59	0.08	0.88	1.44	0.56***
	(1.81)	(1.81)	(0.04)	(1.07)	(0.99)	(0.45)	(1.45)	(1.68)	(0.00)
Consumption over the past 7 days (vegetables)	3.73	5.64	1.92***	4.11	5.56	1.46***	3.99	5.61	1.62***
	(2.93)	(1.66)	(0.00)	(2.75)	(1.78)	(0.00)	(2.81)	(1.71)	(0.00)
Consumption over the past 7 days (fruit)	0.54	0.72	0.18	0.31	0.59	0.27*	0.38	0.66	0.28**
	(1.31)	(1.34)	(0.17)	(1.03)	(1.22)	(0.02)	(1.13)	(1.29)	(0.00)
Consumption over the past 7 days (oil)	4.90	5.79	0.89***	1.99	2.78	0.79***	2.87	4.49	1.62***
	(2.37)	(1.81)	(0.00)	(2.38)	(2.24)	(0.00)	(2.73)	(2.50)	(0.00)
Consumption over the past 7 days (sugar)	2.36	3.53	1.17***	1.30	1.86	0.55**	1.62	2.81	1.18***
	(2.60)	(2.78)	(0.00)	(2.02)	(1.98)	(0.01)	(2.26)	(2.60)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2.2 Reduced Coping Strategies Index (rCSI)

The reduced Coping Strategies index (rCSI) is used to compare households'stress levels due to food shortages. It measures the frequency and severity of the food consumption related behaviours by households to cope with food shortage within a 7 days recall period. The indicator is comprised of 5 standard strategies framed around how and who consumes food in times of shortage. It serves as a proxy indicator for food access.

Table 13 presents the results of the rCSI module for the three sub-samples. The results show that consistently, households reported higher rCSI (2.9 points amongst the phone owners group, 3.95 in the none phone owners and 3.52 in the combined sample respectively). This trend is further reflected in classifying more respondents as being food secure (rCSI severe category). The difference is mainly driven by more households over the phone restricting adult food consumption, reducing meals per day and meal sizes eaten.

Table 12: Impact of Survey Modality on Reduced Coping Strategy Index Classification

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Reduced Coping Strategies Index (rCSI)	14.35	17.27	2.92**	13.16	16.75	3.59***	13.52	17.05	3.52***
	(9.97)	(8.88)	(0.00)	(10.70)	(9.82)	(0.00)	(10.49)	(9.28)	(0.00)
rCSI: Low	0.06	0.04	-0.02	0.19	0.07	-0.13***	0.15	0.05	-0.10***
	(0.25)	(0.21)	(0.36)	(0.40)	(0.25)	(0.00)	(0.36)	(0.23)	(0.00)
rCSI: Median	0.70	0.57	-0.14**	0.55	0.57	0.02	0.60	0.57	-0.03
	(0.46)	(0.50)	(0.00)	(0.50)	(0.50)	(0.66)	(0.49)	(0.50)	(0.41)
rCSI: Severe	0.23	0.39	0.16***	0.26	0.36	0.11*	0.25	0.38	0.13***
	(0.42)	(0.49)	(0.00)	(0.44)	(0.48)	(0.03)	(0.43)	(0.49)	(0.00)
Relied on less preferred, less expensive food	3.92	2.75	-1.17***	3.39	3.29	-0.10	3.55	2.98	-0.57***
	(2.00)	(1.84)	(0.00)	(2.43)	(2.45)	(0.67)	(2.32)	(2.14)	(0.00)
Borrowed food or relied on help from friends or relatives	0.77	0.89	0.12	0.82	0.74	-0.08	0.81	0.82	0.02
	(1.15)	(1.25)	(0.32)	(1.26)	(1.03)	(0.43)	(1.23)	(1.16)	(0.84)
Reduced portion size of meals at meals time	1.98	2.65	0.67**	2.01	2.45	0.44*	2.00	2.57	0.57***
	(2.27)	(2.14)	(0.00)	(2.39)	(2.09)	(0.04)	(2.35)	(2.12)	(0.00)
Restricted consumption by adults in order for young-children to eat	1.73	2.70	0.97***	1.52	2.42	0.90***	1.58	2.58	1.00***
	(1.90)	(1.83)	(0.00)	(1.80)	(2.16)	(0.00)	(1.83)	(1.98)	(0.00)
Reduced the number of meals eaten per day	1.74	2.00	0.26	1.55	2.27	0.72***	1.61	2.12	0.51***
	(2.38)	(2.12)	(0.24)	(2.42)	(2.06)	(0.00)	(2.41)	(2.09)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2.3 Livelihood Coping Strategies (LCS)

The livelihood coping strategy indicator is a household-level indicator that measures the household's capacity to deal with the insufficient food consumption because of shortages or the lack of money in the medium- to long-term bases (30days to 12months) to meet their dietary requirements. Broadly, the strategies are categorized into 3 broad groups: stress, crisis and emergency level. For purposes of targeting and assistance, only the LCS crisis and emergency categories are usually considered (these categories classify households into food insecure).

Table 14 shows the aggregated LCS results categorized into the 4 main categories. The results show some consistency in both magnitude and direction of the difference between F2F and remote, the remote sample consistently reported doing worse compared to the F2F sample (28% more households in the phone-owners sub-sample, 30% in the non-phone owners and 29% in the combined sample cumulatively for crisis and emergency categories). In general, reporting emergency coping was 4 to 5 times higher in the remote compared to F2F across the different sub-groups of analysis.

Table 13: Impact of Survey Modality on Livelihood Coping Strategy

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Maximum Livelihood Coping: None	0.20	0.11	-0.09*	0.39	0.13	-0.26***	0.33	0.12	-0.21***
	(0.40)	(0.32)	(0.01)	(0.49)	(0.34)	(0.00)	(0.47)	(0.33)	(0.00)
Maximum Livelihood Coping: Stress level	0.61	0.43	-0.19***	0.43	0.39	-0.04	0.48	0.41	-0.07*
	(0.49)	(0.50)	(0.00)	(0.50)	(0.49)	(0.39)	(0.50)	(0.49)	(0.03)
Maximum Livelihood Coping: Crisis level	0.14	0.26	0.13**	0.13	0.21	0.08*	0.13	0.24	0.11***
	(0.34)	(0.44)	(0.00)	(0.33)	(0.41)	(0.04)	(0.34)	(0.43)	(0.00)
Maximum Livelihood Coping: Emergency level	0.05	0.20	0.15***	0.05	0.27	0.22***	0.05	0.23	0.18***
	(0.22)	(0.40)	(0.00)	(0.22)	(0.45)	(0.00)	(0.22)	(0.42)	(0.00)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

To understand the composition of the different categories reported in Table 14, the individual strategies adopted are disaggregated and reported in Table 15 below. Among the phone-owners and non-phone owners' sub-samples, strategies such as selling productive assets, scavenging, reduced health care expenditure and selling places of dwelling drive the significant difference between the remote and F2F modality. However, in the full sample, the remote modality significantly adopted more of each strategy than the F2F modality.

Table 14: CAR: Each Coping Strategy Used

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Used Stress coping: Sold household assets/goods	0.28	0.46	0.18***	0.15	0.29	0.14**	0.19	0.38	0.20***
	(0.45)	(0.50)	(0.00)	(0.35)	(0.45)	(0.00)	(0.39)	(0.49)	(0.00)
Used Stress coping: Reduced expenses on health and education	0.23	0.43	0.20***	0.18	0.40	0.23***	0.19	0.42	0.22***
	(0.42)	(0.50)	(0.00)	(0.38)	(0.49)	(0.00)	(0.39)	(0.49)	(0.00)
Used Stress coping: Spent savings	0.47	0.50	0.03	0.22	0.31	0.09	0.30	0.42	0.12***
	(0.50)	(0.50)	(0.50)	(0.42)	(0.46)	(0.06)	(0.46)	(0.49)	(0.00)
Used Stress coping: Borrowed money	0.46	0.53	0.07	0.31	0.34	0.02	0.36	0.45	0.09**
	(0.50)	(0.50)	(0.18)	(0.46)	(0.47)	(0.62)	(0.48)	(0.50)	(0.01)
Used Crisis coping: Sold productive assets or means of transport	0.07	0.26	0.19***	0.02	0.07	0.05*	0.04	0.18	0.14***
	(0.26)	(0.44)	(0.00)	(0.14)	(0.25)	(0.05)	(0.19)	(0.38)	(0.00)
Used Crisis coping: Moved to somewhere else	0.05	0.14	0.09**	0.11	0.24	0.13**	0.09	0.18	0.09***
	(0.23)	(0.35)	(0.01)	(0.31)	(0.43)	(0.00)	(0.29)	(0.39)	(0.00)
Used Crisis coping: Withdrew children from school	0.03	0.05	0.02	0.03	0.09	0.06*	0.03	0.07	0.04*
	(0.17)	(0.22)	(0.32)	(0.17)	(0.29)	(0.02)	(0.17)	(0.25)	(0.01)
Used Emergency coping: Sold houses	0.03	0.14	0.11***	0.02	0.22	0.20***	0.02	0.17	0.15***
	(0.17)	(0.35)	(0.00)	(0.15)	(0.41)	(0.00)	(0.16)	(0.38)	(0.00)
Used Emergency coping: Begged and/or scavenged	0.01	0.07	0.06**	0.02	0.08	0.05*	0.02	0.07	0.05**
	(0.12)	(0.26)	(0.01)	(0.15)	(0.27)	(0.04)	(0.14)	(0.26)	(0.00)
Used Emergency coping: Engaged in illegal income activities	0.01	0.03	0.02	0.01	0.05	0.04*	0.01	0.04	0.03*
	(0.10)	(0.16)	(0.27)	(0.09)	(0.22)	(0.04)	(0.10)	(0.19)	(0.02)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2.4 Food Expenditure Share (FES)

The food expenditure share (FES) is a proxy food security indicator used to measure the economic vulnerability of a household. It is given as the share of household's total consumption expenditure spent on food. The higher the share of food expenditure of a household, the more vulnerable/food insecure the household.

Table 16 presents the summarize FES variable across the different groups. The aggregate FES is ranging from 67 to 73 percentage depending on the whether the respondent is coming from the phone-owners' or non-phone owners' sub-samples. For the phone owners' sample, the remote modality estimates FES at 73 percentage (6 percentage points higher than F2F). The effect is

that, it significantly overestimates the number of category 4 (>= 75%) severely food insecure by 10 percentage points. On the none phone owners' sub-sample, the aggregate FES is significantly lower in remote modality compared to the F2F modality by 6 percentage points reflected in the FES category 4 (underestimating this category by 11 percentage points).

Once the two samples are combined, there is no significant difference between the aggregate FES from remote and F2F modalities.

Table 15: Impact of Survey Modality on Expenditure and Food Expenditure Share

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Expenditure Share	0.67	0.73	0.06*	0.73	0.68	-0.06*	0.71	0.70	-0.01
	(0.20)	(0.24)	(0.01)	(0.17)	(0.25)	(0.02)	(0.18)	(0.24)	(0.52)
FES < 50 percent	0.20	0.16	-0.04	0.10	0.25	0.15***	0.13	0.20	0.07*
	(0.40)	(0.37)	(0.27)	(0.30)	(0.44)	(0.00)	(0.34)	(0.40)	(0.01)
FES: 50-65 percent	0.23	0.17	-0.06	0.16	0.18	0.02	0.18	0.17	-0.00
	(0.42)	(0.38)	(0.15)	(0.36)	(0.38)	(0.61)	(0.38)	(0.38)	(0.86)
FES: 65-75 percent	0.13	0.13	-0.01	0.17	0.11	-0.06	0.16	0.12	-0.04
	(0.34)	(0.33)	(0.87)	(0.38)	(0.31)	(0.06)	(0.37)	(0.33)	(0.08)
FES: >= 75 percent	0.44	0.54	0.10*	0.57	0.46	-0.11*	0.53	0.51	-0.02
	(0.50)	(0.50)	(0.04)	(0.50)	(0.50)	(0.03)	(0.50)	(0.50)	(0.50)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

4.2.5 Income Source and Income Change

The income indicator captures the respondent/household income source and income change experienced in the last 12 months prior to the interview. This indicator is a recent additionwas introduced to calculate CARI in remote assessment during COVID 19 and it is not yet part of the to the CARI console. The indicator to reflects the economic situation of the respondent/household in the absence of the FES. This was especially important during the COVID-19 outbreak when exigencies of the time did not allow for the collection of expenditure to calculate FES. The type of income generating activity and how economic situation at the time impacted on income flow either positive or negative was considered a good proxy for vulnerable the household could be. Based on the type of income (regular/irregular) and the change to income (change/no change) to the income stream, the respondent/household is classified into one of 4 CARI categories (see Table 17 for the categories). The irregular & reduced, and no income sources groups corresponding to FES category 3 (FES: 65-75%) and category 4 (FES: >= 75%), respectively.

The results of the income source and income change indicator comparison over modality shows that, respondents tend to underreport their food security over the phone (more households reported stable regular income source in the F2F interviews than remote across all the sub-samples). Although this may be driven by the categorization of agricultural income as regular income, the effect should be same across modalities. On the other hand, respondents reported engaging in significantly more irregular income sources (unstable and reducing income) over the phone compared to F2F.

Table 16: Impact of Survey Modality on Income

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Income: Regular and Unchanged/Increasing	0.45	0.26	-0.19***	0.44	0.24	-0.19***	0.44	0.25	-0.19***
	(0.50)	(0.44)	(0.00)	(0.50)	(0.43)	(0.00)	(0.50)	(0.44)	(0.00)
Income: Regular and Reduced OR Irregular and Unchanged/Increasing	0.42	0.50	0.08	0.46	0.49	0.02	0.45	0.49	0.04
	(0.49)	(0.50)	(0.13)	(0.50)	(0.50)	(0.63)	(0.50)	(0.50)	(0.22)
Income: Irregular and Reduced	0.12	0.23	0.11**	0.08	0.25	0.17***	0.09	0.24	0.15***
	(0.32)	(0.42)	(0.00)	(0.27)	(0.44)	(0.00)	(0.29)	(0.43)	(0.00)
Income: No income sources	0.01	0.01	-0.00	0.02	0.02	-0.00	0.02	0.01	-0.00
	(0.12)	(0.11)	(0.94)	(0.14)	(0.13)	(0.85)	(0.13)	(0.12)	(0.71)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

5 Summary and conclusions

Disclaimer: the conclusions should be interpreted considering the following limitations. First, the final remote sample that has been interviewed is relatively small. Second, although there is no statistical difference between the respondents who received phones from the project and phone owners, their response may be influenced by receipt of the mobile phones. Third, the results have not been weighted to adjust for the different response rates.

The following are some key highlights from the results of the analysis from the CAR pilot study:

- The low phone ownership (25%) and mobile coverage influenced greatly the study implementation, sampling and number of households reached over the phone (remote sample). The response rate was generally low in the remote modality even though phones were distributed to potential respondents.
- Respondents over the phone generally tend to overreport their food security status compared to F2F interviews.
- The results of the food insecurity aggregation show that, rCARI-Inc methodology significantly underestimates acute food insecurity (AFI) by between 9 and 21 percentage points.
- While there is no significant difference between CARI-FES and rCARI-FES in the phone owner sub-sample, rCARI-Inc significantly underestimates food insecurity by up to 7 percentage points.
- In terms of the food security indicators, households interviewed over the phone (remote modality) reports higher food consumption (FCS), worse coping situations (rCSI and LCS) compared to when interviewed in F2F. Also, when interviewed remotely, households reported significantly higher FES compared to F2F.
- The analysis of the long- and short- LCS, FES and FCS modules (F2F) on the whole does not show any statistically significant difference. This means that by fine-tuning the short modules could potentially replace the longer versions in F2F surveys to reduce survey fatigue both for the interviewee and interviewer.
- The results comparing the standard F2F sub-sample with the phone owners sub-sample (typical CATI sample) shows that using the standard CATI in regions/countries/areas where phone

penetration is low has a tendency to exclude more vulnerable and food insecure households in so doing underreporting AFI.

5.1 Study implementation challenges in CAR

Although during the field work, extensive efforts were made to reduce and manage implementation challenges, a number of these remained and affected the successful implementation of the survey. These include:

1. Restricted access to the field because of deteriorating security situation
2. Poor mobile phone network/infrastructure

a. Most parts of the country do not have mobile phone network coverage.

b. In places where mobile coverage exists, poor infrastructure and frequent power cuts affected call quality.

c. Access to electricity for charging phones particularly in rural areas.

3. Low phone ownership in the population

a. Only 25% of households in the sample owned phones.

4. Logistical challenges.

a. Acquisition and distribution of mobile phones/SIM cards to study sample without phones.

b. Had to rely on UNHAS (United Nations Humanitarian Air Service) for enumerator deployment affecting timely start of the survey.

5. Low response rate in the remote arm (High Attrition).
6. Lack of an established platform for remote data collection.
7. Weather disruptions/rainy season

6 Regression analysis

In this section of the analysis, regression analysis are conducted to determine the role of phone ownership and modality of interview administration controlling individual and household level covariates.

6.1 Next Steps...

1. Conduct further in-depth analysis of the data - exploring the association between the indicators especially FES, income indicators and other statistical diagnostic analysis.
2. Apply weights where practicable to address the issues of lower response rates in the remote sample.
3. Prepare the draft manuscripts for peer review publication.

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8 Annexes

8.1 Final survey sample size & and phone access by ADMIN area

Table 17: Listed Sample: ADMIN Area Selection and Distribution of Phone Ownership

	Household owns phone		
	No	Yes	Total
N	1,452 (69.8%)	628 (30.2%)	2,080 (100.0%)
Geographic Region: Admin 1			
Mbomou	260 (17.9%)	425 (67.7%)	685 (32.9%)
Nana-Gribizi	1,192 (82.1%)	203 (32.3%)	1,395 (67.1%)
Geographic Region: Admin 2			
Bangassou	260 (17.9%)	425 (67.7%)	685 (32.9%)
Kaga-Bandaro	1,192 (82.1%)	203 (32.3%)	1,395 (67.1%)
Geographic Region: Admin 3			
Nana	467 (32.2%)	51 (8.1%)	518 (24.9%)
Botto	356 (24.5%)	18 (2.9%)	374 (18.0%)
Kaga-Bandoro	369 (25.4%)	134 (21.3%)	503 (24.2%)
Bangassou	260 (17.9%)	425 (67.7%)	685 (32.9%)
Geographic Region: Admin 5			
DAMAKA 2	57 (3.9%)	0 (0.0%)	57 (2.7%)
BISSEKEBOU (1,2,3 et	61 (4.2%)	3 (0.5%)	64 (3.1%)
KIA 3	97 (6.7%)	3 (0.5%)	100 (4.8%)
OUANDAGO 2	83 (5.7%)	15 (2.4%)	98 (4.7%)
YAFARA - GOUVA	77 (5.3%)	23 (3.7%)	100 (4.8%)
KONVI 1	92 (6.3%)	7 (1.1%)	99 (4.8%)
ZEFIO	92 (6.3%)	4 (0.6%)	96 (4.6%)
BEGUEDE 2	98 (6.7%)	2 (0.3%)	100 (4.8%)
BOLAFEI	78 (5.4%)	0 (0.0%)	78 (3.8%)
KPOKPO 1	88 (6.1%)	12 (1.9%)	100 (4.8%)
MISSION CATHOLIQUE	30 (2.1%)	67 (10.7%)	97 (4.7%)
MBROUA 2	70 (4.8%)	34 (5.4%)	104 (5.0%)
GRIBINGUI 1	103 (7.1%)	8 (1.3%)	111 (5.3%)
KPOPO 2	88 (6.1%)	12 (1.9%)	100 (4.8%)
GRIBINGUI 2	78 (5.4%)	13 (2.1%)	91 (4.4%)
KOBO	25 (1.7%)	54 (8.6%)	79 (3.8%)
GBIAKOTA	22 (1.5%)	44 (7.0%)	66 (3.2%)
MALIKO 2	30 (2.1%)	45 (7.2%)	75 (3.6%)
DAKPOUTOU	30 (2.1%)	50 (8.0%)	80 (3.8%)
AOUYA	17 (1.2%)	53 (8.4%)	70 (3.4%)
BEREME	21 (1.4%)	7 (1.1%)	28 (1.3%)
KEMBE-LIGNE	34 (2.3%)	31 (4.9%)	65 (3.1%)
SAOZANGANDOU	29 (2.0%)	43 (6.8%)	72 (3.5%)
LIOUANGO-TIRAILLEUR	21 (1.4%)	60 (9.6%)	81 (3.9%)
SAMBA-GAMADE	31 (2.1%)	38 (6.1%)	69 (3.3%)

8.2 F2F Attrition

Table 18: Balance Table for Randomized Questionnaire Length Assignment in F2F Survey: Completed vs. Attrition

	Own Phone			No Phone			Total F2F		
Variable	Completed	Attrition	Diff	Completed	Attrition	Diff	Completed	Attrition	Diff
List respondent age	36.27	32.65	-3.61	39.47	38.89	-0.59	38.50	37.05	-1.46
	(12.55)	(10.44)	(0.09)	(13.82)	(15.33)	(0.77)	(13.52)	(14.30)	(0.36)
Household head is the respondent	0.72	0.62	-0.11	0.79	0.68	-0.11	0.77	0.66	-0.11*
	(0.45)	(0.50)	(0.27)	(0.41)	(0.47)	(0.08)	(0.42)	(0.48)	(0.04)
Female Household Head	0.19	0.27	0.08	0.30	0.31	0.01	0.26	0.30	0.03
	(0.39)	(0.45)	(0.38)	(0.46)	(0.46)	(0.87)	(0.44)	(0.46)	(0.54)
Married Household Head	0.27	0.12	-0.15*	0.14	0.10	-0.05	0.18	0.10	-0.08*
	(0.44)	(0.33)	(0.03)	(0.35)	(0.30)	(0.24)	(0.38)	(0.30)	(0.02)
Household head years of education	7.55	7.00	-0.55	3.41	3.16	-0.25	4.66	4.30	-0.37
	(4.13)	(3.77)	(0.48)	(3.25)	(3.52)	(0.59)	(4.02)	(3.98)	(0.40)
Displaced household	0.09	0.27	0.18*	0.03	0.06	0.03	0.05	0.12	0.08*
	(0.29)	(0.45)	(0.05)	(0.17)	(0.25)	(0.30)	(0.22)	(0.33)	(0.04)
Household size	7.72	7.23	-0.49	6.08	6.23	0.15	6.57	6.52	-0.05
	(3.44)	(4.18)	(0.56)	(2.67)	(2.58)	(0.66)	(3.02)	(3.15)	(0.88)
Number of household member younger than 15	4.04	3.77	-0.27	3.23	2.98	-0.25	3.48	3.22	-0.26
	(2.27)	(2.39)	(0.57)	(1.95)	(1.49)	(0.22)	(2.08)	(1.82)	(0.20)
Number of household member older than 60	0.21	0.19	-0.02	0.17	0.18	0.00	0.19	0.18	-0.00
	(0.49)	(0.40)	(0.82)	(0.44)	(0.39)	(0.96)	(0.46)	(0.39)	(0.93)
Has household member with disability/chronic disease	0.30	0.27	-0.03	0.15	0.19	0.05	0.19	0.22	0.02
	(0.46)	(0.45)	(0.77)	(0.35)	(0.40)	(0.37)	(0.39)	(0.41)	(0.60)
Number of rooms	2.89	2.27	-0.62*	1.99	1.66	-0.33*	2.26	1.84	-0.42***
	(1.24)	(1.22)	(0.01)	(1.08)	(1.05)	(0.02)	(1.20)	(1.13)	(0.00)
Household Crowding Index	3.02	3.77	0.74	3.93	4.97	1.04*	3.66	4.62	0.96**
	(1.70)	(2.62)	(0.15)	(2.87)	(3.50)	(0.02)	(2.61)	(3.30)	(0.01)
Household has impoverished roof	0.68	0.54	-0.14	0.85	0.84	-0.01	0.80	0.75	-0.05
	(0.47)	(0.51)	(0.17)	(0.36)	(0.37)	(0.86)	(0.40)	(0.44)	(0.34)
Household has impoverished floor	0.90	0.77	-0.13	0.96	0.95	-0.01	0.94	0.90	-0.05
	(0.30)	(0.43)	(0.13)	(0.19)	(0.22)	(0.69)	(0.23)	(0.30)	(0.17)
Observations	294	26	320	675	62	737	969	88	1,057

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

8.3 Remote Attrition

Table 19: Balance Table for Randomized Questionnaire Length Assignment in Remote Survey: Completed vs. Attrition

	Own Phone			No Phone			Total Remote		
Variable	Completed	Attrition	Diff	Completed	Attrition	Diff	Completed	Attrition	Diff
List respondent age	39.03	36.09	-2.95*	39.76	39.27	-0.49	39.34	38.62	-0.72
	(12.38)	(13.01)	(0.04)	(13.94)	(14.31)	(0.73)	(13.05)	(14.11)	(0.44)
Household head is the respondent	0.78	0.68	-0.09	0.82	0.79	-0.03	0.80	0.77	-0.03
	(0.42)	(0.47)	(0.06)	(0.38)	(0.41)	(0.37)	(0.40)	(0.42)	(0.30)
Female Household Head	0.25	0.21	-0.05	0.32	0.31	-0.01	0.28	0.29	0.00
	(0.44)	(0.41)	(0.30)	(0.47)	(0.46)	(0.77)	(0.45)	(0.45)	(0.94)
Married Household Head	0.27	0.25	-0.03	0.14	0.10	-0.04	0.22	0.13	-0.08**
	(0.45)	(0.43)	(0.56)	(0.35)	(0.31)	(0.26)	(0.41)	(0.34)	(0.00)
Household head years of education	7.92	7.03	-0.89	3.54	3.26	-0.28	6.03	4.02	-2.01***
	(4.69)	(4.26)	(0.08)	(3.49)	(3.17)	(0.42)	(4.73)	(3.74)	(0.00)
Displaced household	0.15	0.08	-0.07*	0.03	0.03	-0.00	0.10	0.04	-0.06**
	(0.36)	(0.27)	(0.04)	(0.18)	(0.18)	(0.92)	(0.30)	(0.20)	(0.00)
Household size	8.39	7.50	-0.89	5.99	5.96	-0.04	7.36	6.27	-1.09***
	(3.40)	(4.42)	(0.05)	(3.05)	(2.57)	(0.91)	(3.46)	(3.10)	(0.00)
Number of household member younger than 15	4.31	3.68	-0.63**	3.13	3.13	-0.00	3.80	3.24	-0.56***
	(2.03)	(2.17)	(0.01)	(2.29)	(1.81)	(1.00)	(2.22)	(1.90)	(0.00)
Number of household member older than 60	0.21	0.24	0.03	0.13	0.19	0.05	0.18	0.20	0.02
	(0.44)	(0.64)	(0.65)	(0.41)	(0.49)	(0.21)	(0.43)	(0.53)	(0.52)
Has household member with disability/chronic disease	0.31	0.28	-0.02	0.15	0.17	0.01	0.24	0.19	-0.05
	(0.46)	(0.45)	(0.69)	(0.36)	(0.37)	(0.68)	(0.43)	(0.39)	(0.10)
Number of rooms	3.13	2.85	-0.29	1.95	1.91	-0.04	2.62	2.10	-0.53***
	(1.27)	(1.40)	(0.06)	(1.09)	(1.09)	(0.69)	(1.33)	(1.22)	(0.00)
Household Crowding Index	2.97	3.04	0.06	3.80	4.04	0.25	3.33	3.84	0.51**
	(1.26)	(1.97)	(0.74)	(2.68)	(3.02)	(0.37)	(2.04)	(2.87)	(0.00)
Household has impoverished roof	0.63	0.68	0.05	0.86	0.86	0.00	0.73	0.82	0.09**
	(0.48)	(0.47)	(0.34)	(0.35)	(0.35)	(0.99)	(0.45)	(0.38)	(0.00)
Household has impoverished floor	0.92	0.90	-0.02	0.97	0.96	-0.01	0.94	0.95	0.01
	(0.28)	(0.30)	(0.62)	(0.16)	(0.19)	(0.42)	(0.23)	(0.22)	(0.66)
Observations	157	151	308	119	596	715	276	747	1,023

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

8.4 Supporting documents

8.4.1 CARI resources

- CARI methodology Consolidated Approach for Reporting Indicators of Food Security (CARI) Guidelines - Third edition
- FCS
- rCSI
- LCS
- FES

8.4.2 Listing questionnaire

	Own Phone			No Phone			Total		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
List respondent age	36.27	39.03	2.76*	39.47	39.76	0.28	38.50	39.34	0.84
	(12.55)	(12.38)	(0.03)	(13.82)	(13.94)	(0.84)	(13.52)	(13.05)	(0.35)
Household head is the respondent	0.72	0.78	0.05	0.79	0.82	0.04	0.77	0.80	0.03
	(0.45)	(0.42)	(0.21)	(0.41)	(0.38)	(0.34)	(0.42)	(0.40)	(0.29)
Female Household Head	0.19	0.25	0.06	0.30	0.32	0.02	0.26	0.28	0.02
	(0.39)	(0.44)	(0.12)	(0.46)	(0.47)	(0.62)	(0.44)	(0.45)	(0.55)
Married Household Head	0.27	0.27	0.01	0.14	0.14	-0.00	0.18	0.22	0.04
	(0.44)	(0.45)	(0.85)	(0.35)	(0.35)	(0.98)	(0.38)	(0.41)	(0.19)
Household head years of education	7.55	7.92	0.38	3.41	3.54	0.13	4.66	6.03	1.37***
	(4.13)	(4.69)	(0.40)	(3.25)	(3.49)	(0.70)	(4.02)	(4.73)	(0.00)
Displaced household	0.09	0.15	0.06	0.03	0.03	0.00	0.05	0.10	0.05**
	(0.29)	(0.36)	(0.07)	(0.17)	(0.18)	(0.89)	(0.22)	(0.30)	(0.01)
Household size	7.72	8.39	0.67*	6.08	5.99	-0.08	6.57	7.36	0.78***
	(3.44)	(3.40)	(0.05)	(2.67)	(3.05)	(0.78)	(3.02)	(3.46)	(0.00)
Number of household member younger than 15	4.04	4.31	0.27	3.23	3.13	-0.10	3.48	3.80	0.32*
	(2.27)	(2.03)	(0.20)	(1.95)	(2.29)	(0.64)	(2.08)	(2.22)	(0.03)
Number of household member older than 60	0.21	0.21	-0.00	0.17	0.13	-0.04	0.19	0.18	-0.01
	(0.49)	(0.44)	(0.99)	(0.44)	(0.41)	(0.33)	(0.46)	(0.43)	(0.78)
Has household member with disability/chronic disease	0.30	0.31	0.01	0.15	0.15	0.00	0.19	0.24	0.05
	(0.46)	(0.46)	(0.83)	(0.35)	(0.36)	(0.90)	(0.39)	(0.43)	(0.10)
Number of rooms	2.89	3.13	0.24	1.99	1.95	-0.04	2.26	2.62	0.36***
	(1.24)	(1.27)	(0.05)	(1.08)	(1.09)	(0.72)	(1.20)	(1.33)	(0.00)
Household Crowding Index	3.02	2.97	-0.05	3.93	3.80	-0.13	3.66	3.33	-0.33*
	(1.70)	(1.26)	(0.73)	(2.87)	(2.68)	(0.62)	(2.61)	(2.04)	(0.03)
Household has impoverished roof	0.68	0.63	-0.05	0.85	0.86	0.01	0.80	0.73	-0.07*
	(0.47)	(0.48)	(0.33)	(0.36)	(0.35)	(0.78)	(0.40)	(0.45)	(0.02)
Household has impoverished floor	0.90	0.92	0.02	0.96	0.97	0.01	0.94	0.94	-0.00
	(0.30)	(0.28)	(0.50)	(0.19)	(0.16)	(0.46)	(0.23)	(0.23)	(0.94)
Observations	294	157	451	675	119	794	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 20: Cross Tabulation of Food Insecurity Binary Variables using FES and Income

@ | Y Y Y Y Y Y Y Y Y @

	Food Insecure Group								
	0			*1*			*Total*		
	No.	Col %	Cum %	No.	Col %	Cum %	No.	Col %	Cum %
Food Insecure Group									
0	181	90.0	90.0	239	22.9	22.9	420	33.7	33.7
1	20	10.0	100.0	805	77.1	100.0	825	66.3	100.0
Total	201	100.0		1044	100.0		1245	100.0	